



CLASSIFICATION OF COTTON PLANT DISEASES AND PESTS USING THE CONVOLUTIONAL NEURAL NETWORK METHOD

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Introduction

Global Production of cotton
2022/2023 approximately

118,3 million bales

- The importance of cotton for the industry makes cotton cultivation highly prioritized.
- A challenge in cotton cultivation is the attack of diseases and pests.
- The majority of farmers still perform identification manually.

the limitations of manual disease and pest identification in cotton plants, including:

- **Time-consuming:** Manual identification can be a laborious process, especially for large-scale operations.
- **Inaccurate:** Human judgment can be subjective and prone to errors, leading to misidentification of diseases and pests.
- **Suboptimal management practices:** Incorrect identification can result in ineffective pesticide use, which can harm both the environment and cotton yields.

Introduction

Advances in agricultural technology offer new solutions, one of which is the use of machine learning

The application of Machine Learning technology, namely Convolutional Neural Network (CNN), can perform automatic classification.

This research will classify cotton plant diseases and pests using the CNN method with cotton leaf images as input.

Methodology

- **Datasets**

Diseases



*Bacterial
Blight*



*Powdery
Mildew*



Target Spot

Pests



Aphids



*Army
Worm*

Normal



Healthy



Data Preprocessing

Data Augmentation

Building CNN Model

Model Training

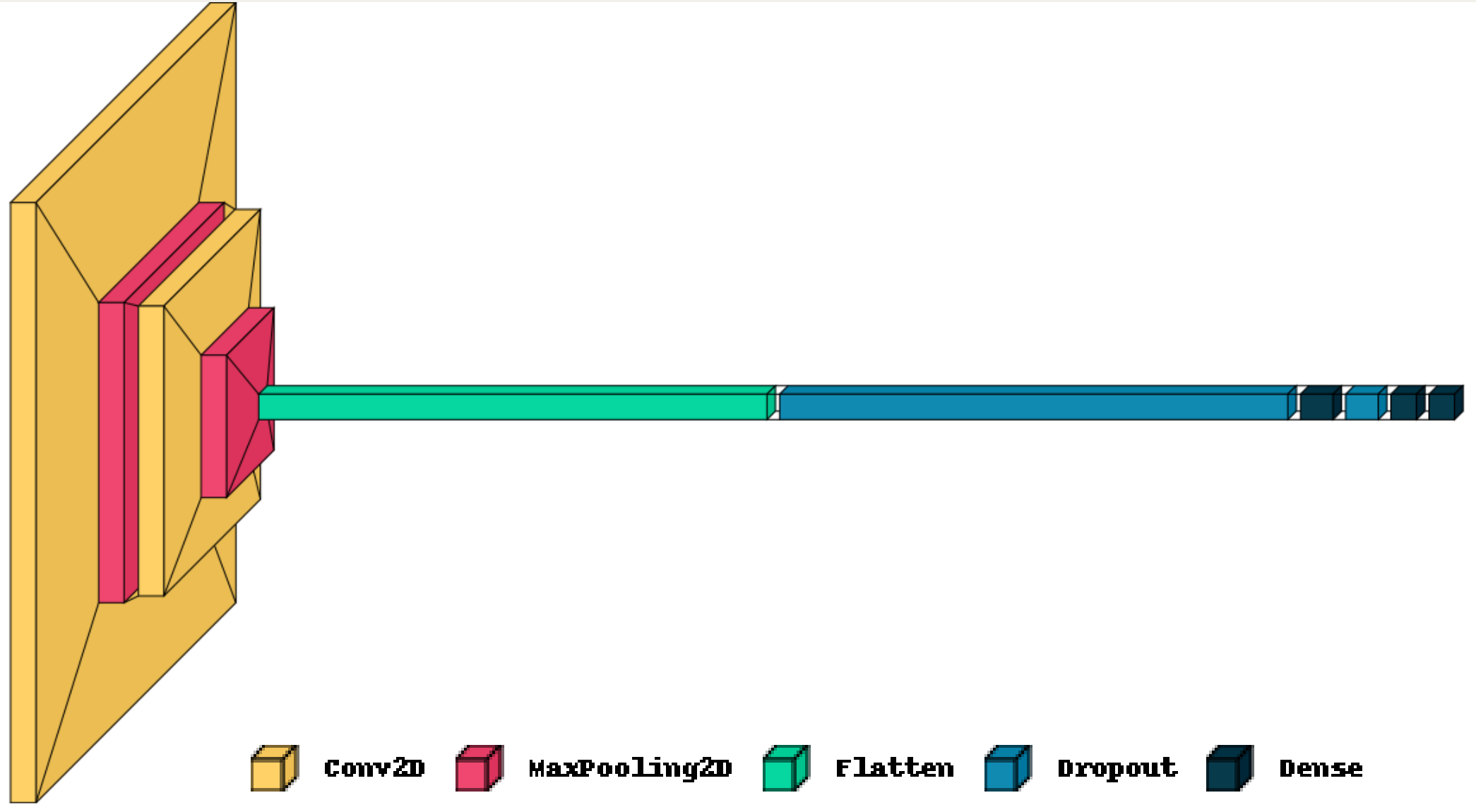
Model Evaluation

- Resize, Normalization

- Shear, Zoom, Rotation, Horizontal flip

Methodology

- Architecture Model

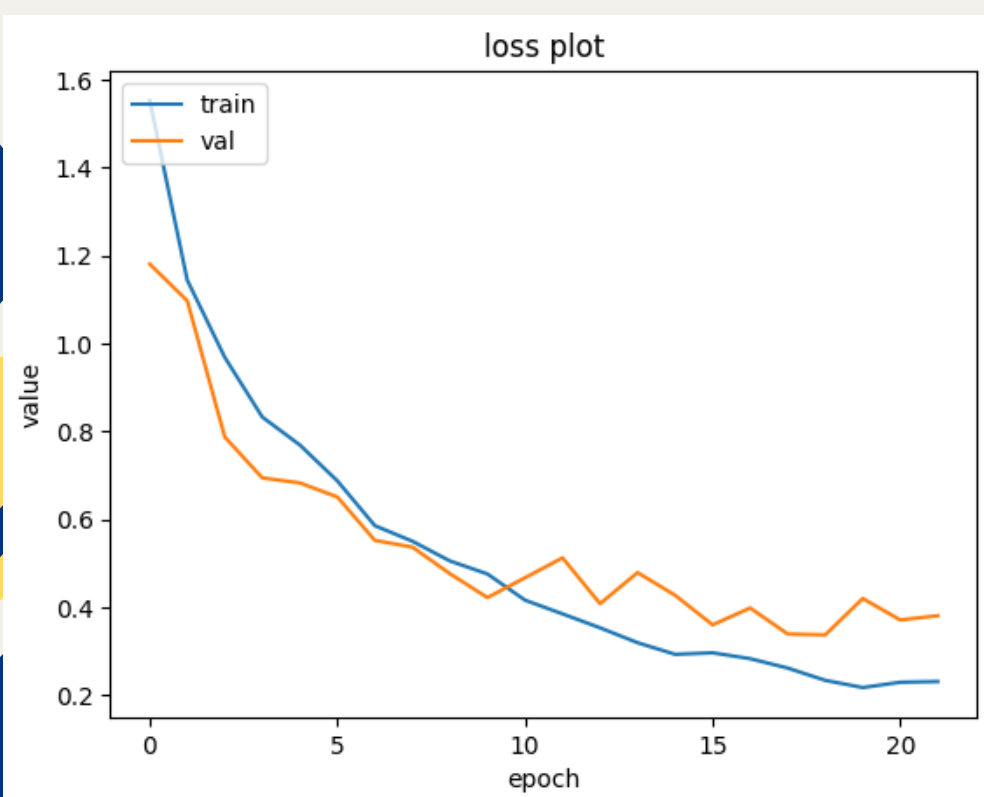
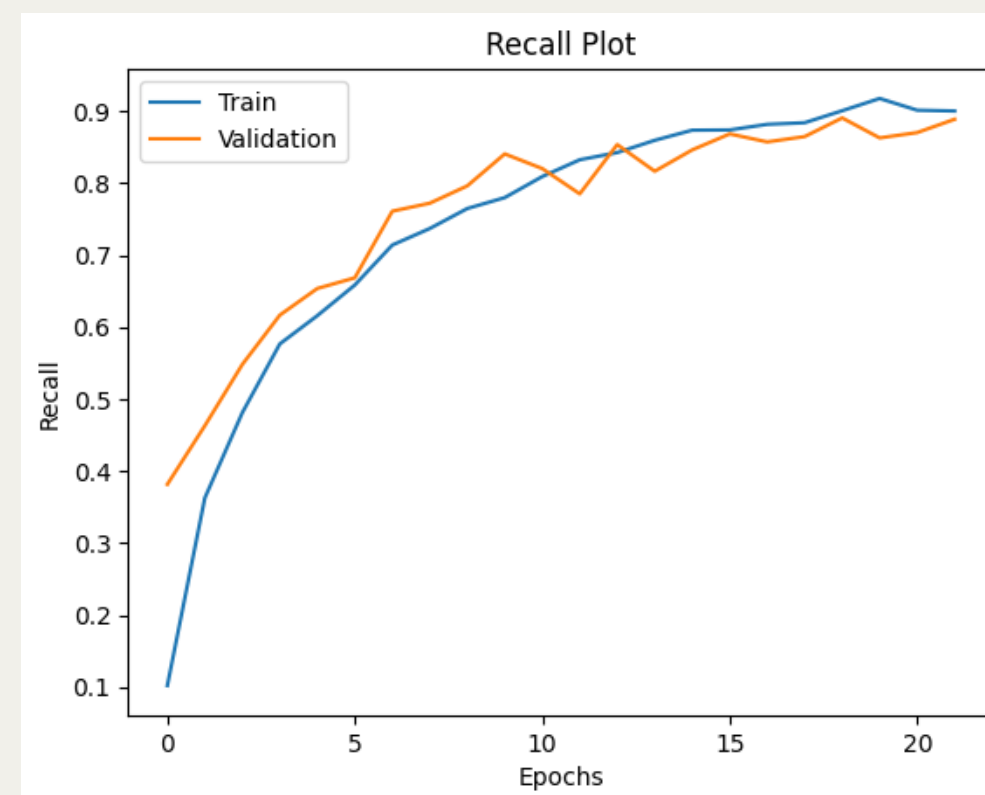
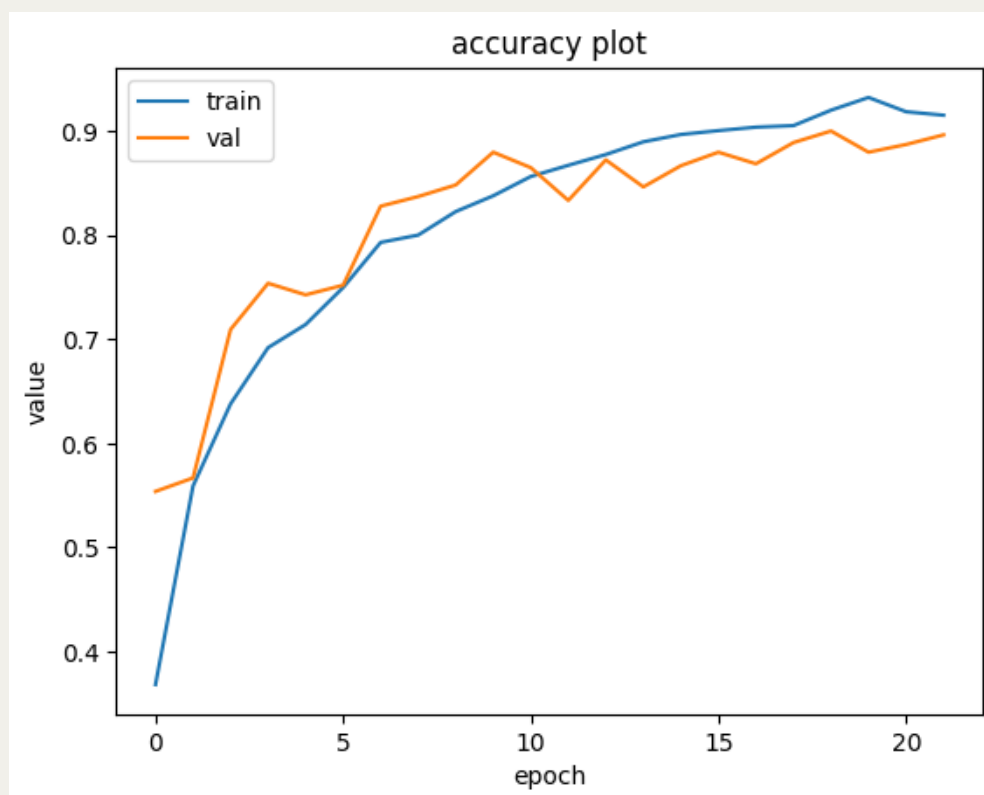


- Data Augmentation

No.	Augmentation	
1.	<i>Shear</i>	 
	Tilting the image	
2	<i>Rotation</i>	 
	Rotating the image by 20 degrees	
3	<i>Zoom</i>	 
	Enlarging the image by 0.2	
4	<i>Horizontal flip</i>	 
	Perform horizontal flip	

Result

- Performance evaluation of the trained CNN model



Epoch 22/25
169/169 [=====] - 129s 762ms/step

loss: 0.2308 accuracy: 0.9152 - recall: 0.9004

val_loss: 0.3805 - val_accuracy: 0.8963 - val_recall: 0.8889

Training Process:

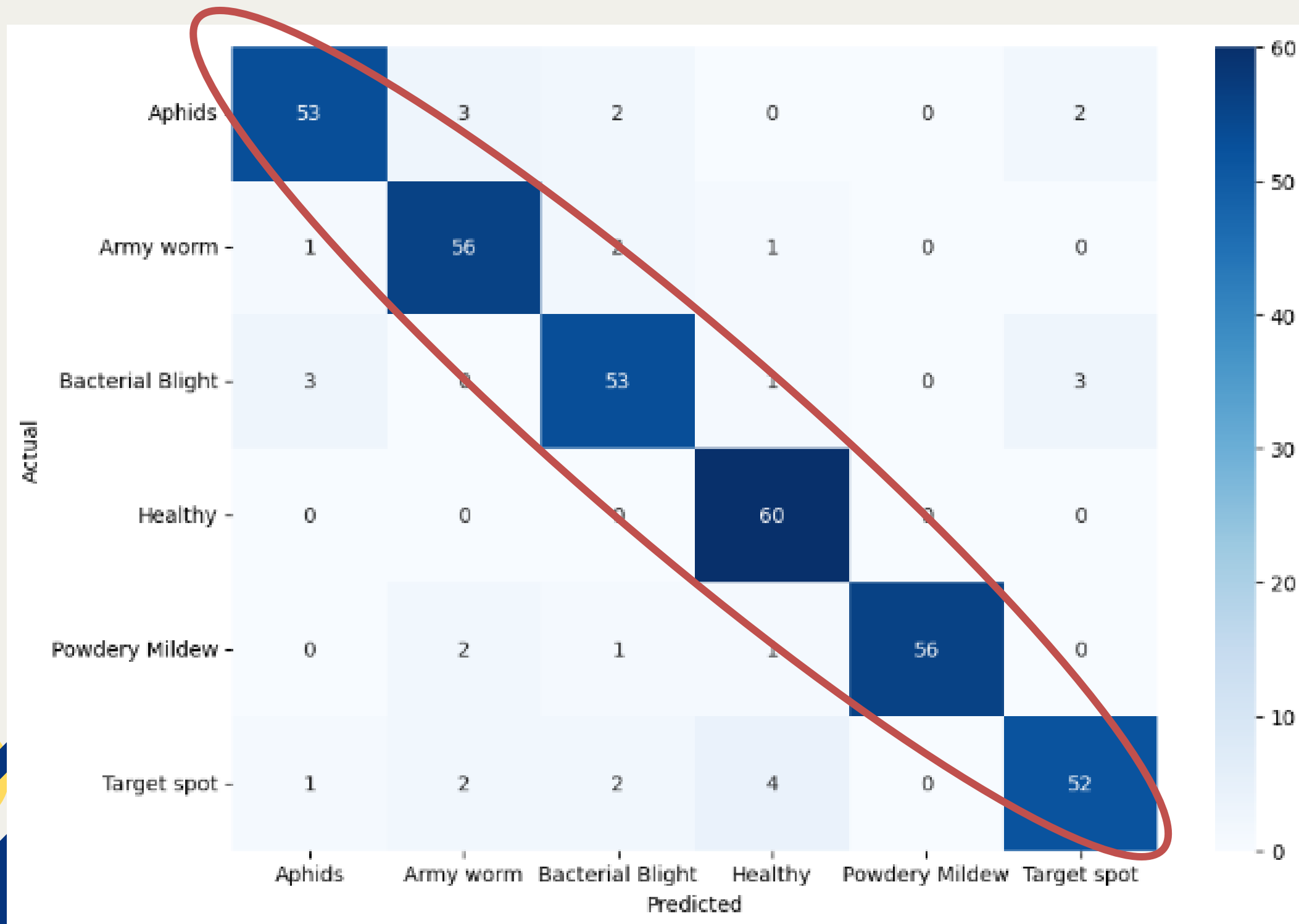
- Trained on train and validation datasets for 22 iterations.
- Entire dataset was trained 22 times.

Evaluation Metrics:

- Accuracy: 91% on training data, 89% on validation data.
- Recall: 90% on training data, 88% on validation data.

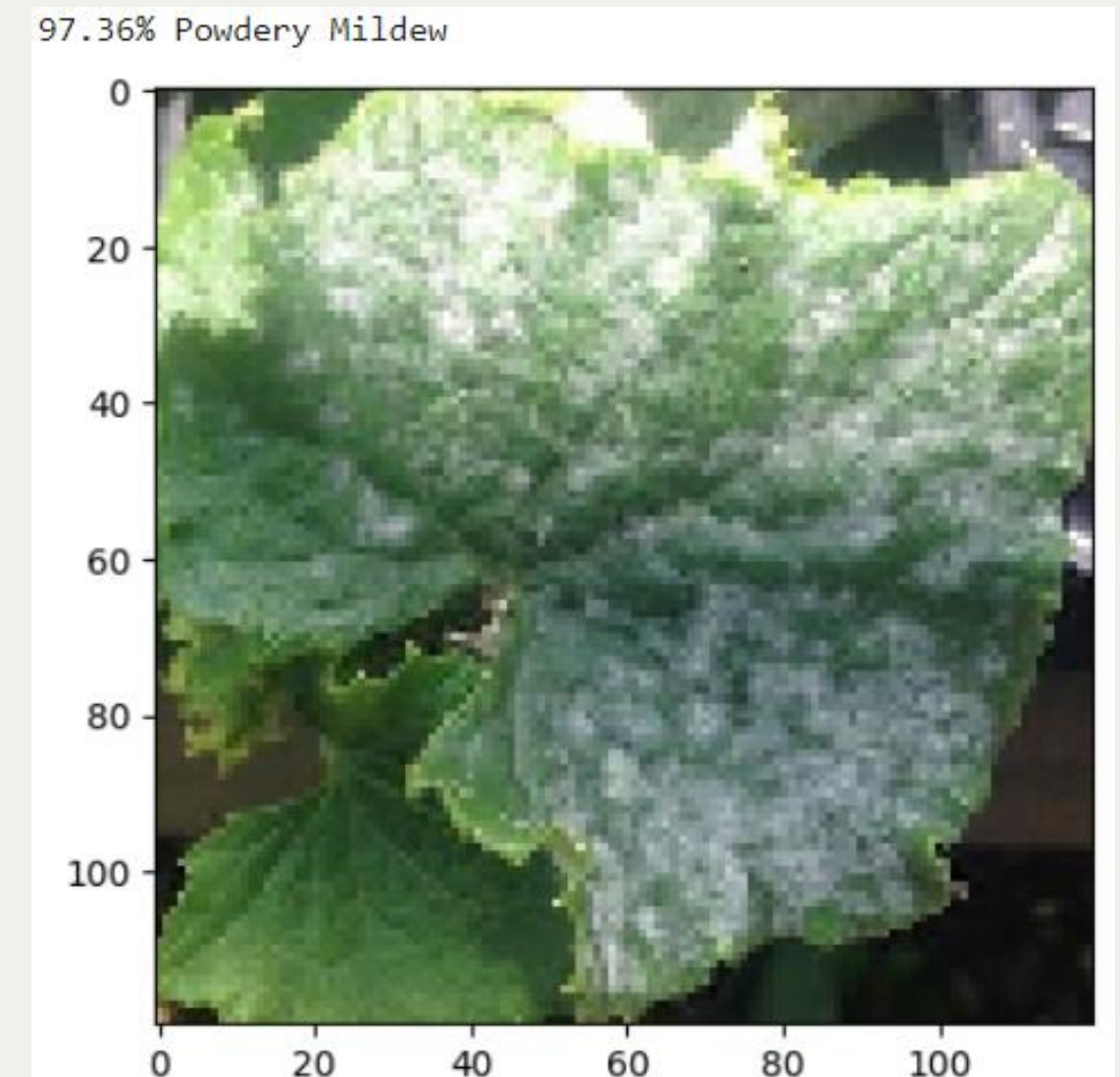
Result

- Test evaluation Model CNN



Testing Metrics:

- Accuracy: 91%
- 330 out of 361 images are predicted correctly



Result

- Confusion Matrices Model CNN

Class	Total Images	TP	TN	FN	FP	Precision	Recall	F1 score
Aphids	60	53	296	7	5	0,91	0,88	0,90
Army Worm	60	56	294	4	7	0,89	0,93	0,91
Bacterial Blight	60	53	294	7	7	0,88	0,88	0,88
Healthy	60	60	294	0	7	0,90	1,00	0,94
Powdery Mildew	60	56	301	4	0	1,00	0,93	0,97
Target spot	61	52	295	9	5	0,91	0,85	0,88

Testing Metrics:

- Accuracy: 91%
- 330 out of 361 images are predicted correctly

Conclusion

- The proposed CNN model successfully achieved high accuracy (91%) and recall (90%) in identifying cotton diseases and pests.
- The model's architecture, composed of feature extraction and classification stages, proved effective for this task.
- The model demonstrated strong generalization capabilities, accurately classifying new data with 91% accuracy.

Convolutional Neural Networks (CNNs) show significant promise for applications in the agricultural industry, specifically for real-time disease and pest classification.

The background features abstract, overlapping shapes in dark blue and yellow. These shapes are primarily located in the corners and along the left edge, creating a modern, geometric aesthetic. The shapes include circles, elongated ovals, and irregular polygons.

THANK YOU

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